Item 03: polynomial-method solving on descended Semaev systems

Verdict

INCONCLUSIVE (level: not reached; the frozen rule's controls and real runs did not execute). b03 summarize on the recorded outputs prints "a control did not pass or is missing: INCONCLUSIVE" and "VERDICT: INCONCLUSIVE (level: n/a)" (raw/summary-partial.txt). Arm (e) msolve remains BLOCKED (not installed). This is not a loss and not a win: only card 01 ran. Cards 02 to 09 were refused by task_card.py before any command started (see Deviations), so no real FES, real dinur/lptwy, positive control 1, negative control cross-check, or F4 run exists.

What did run (card 01, receipt `raw/receipt-card-01.json`)

* Built b03 and b05 (rustc, opt-level 3, target-cpu=native).
* selftest SELFTEST OK: 189 counted-model == real-counter checks, 126 parity checks equal brute force (48 odd), parity check can fail (29 of 60 wrong at t=1).
* asym ASYM OK: Dinur d=2 0.694550 (published 0.6943), d=3 0.811490 (0.8114), lptwy d=2 0.876250 (0.8765); all inside the frozen intervals. Positive control 2 passed.
* sens SENS OK (E=33 FES ratio 1.888, expected 1.889; E=64 3.610 vs 3.611; dinur change 1.4% and 4.0%: exceeds the protocol's 1%, see caveats).
* Counted model for ten cells (targets 0-7 and 8 random systems each).

Counted-model numbers (descended target 6, log2 bit operations; model only)

cell	N	E	FES 2^N(2E+2)	dinur	lptwy	rho_dinur	rho_lptwy
c13-6-2	12	13	16.81	24.93	25.44	0.0036	0.0025
c17-8-2	16	17	21.17	28.47	29.08	0.0063	0.0041
c21-10-2	20	21	25.46	32.47	32.76	0.0077	0.0063
c25-12-2	24	25	29.70	36.11	36.91	0.0118	0.0068
c29-14-2	28	29	33.91	39.82	40.71	0.0166	0.0089
c33-16-2	32	33	38.09	43.64	44.70	0.0214	0.0102

The cost is a function of (N, E, d) only, so the per-cell interval is degenerate (negative-control identity is seen in the logs but not yet checked against real random-system runs). Between N=28 and N=32 the local slopes are dinur 0.955, lptwy 0.998, FES 1.045 per variable: the polynomial-method arms grow more slowly than the Gray-code FES in this range yet remain about 47x (dinur) and 98x (lptwy) more expensive at N=32. Were real runs to confirm this, the rule's branch would be TRADES or REGRESSES depending on the jackknifed fit; it is not applied here because the fit needs summarize rows from the real runs (it printed "fewer than 4 cells with FES rows: no fit"). Crossover N* is not computed (the crossover block is empty without the run rows).

S_4 (cubic) cells, rho_dinur against the fes-floor: 1.7e-3 (N=13), 2.7e-3 (19), 5.1e-3 (25), 6.2e-3 (31).

Exponent statement (made whatever the verdict; level exponent)

Published-form asymptotic c: 0.6946 for d=2 and 0.8115 for d=3. Required for a decomposition win over rho is c < 1/2 - 1/m: 0 for m=2 (distance remaining 0.6946), 1/6 for m=3 (distance remaining 0.6448). Index-calculus exponent relative to rho, 2/m + 2c: 2.389 (m=2) and 2.290 (m=3); above 1 loses. This is an asymptotic-formula statement, not a measured improvement.

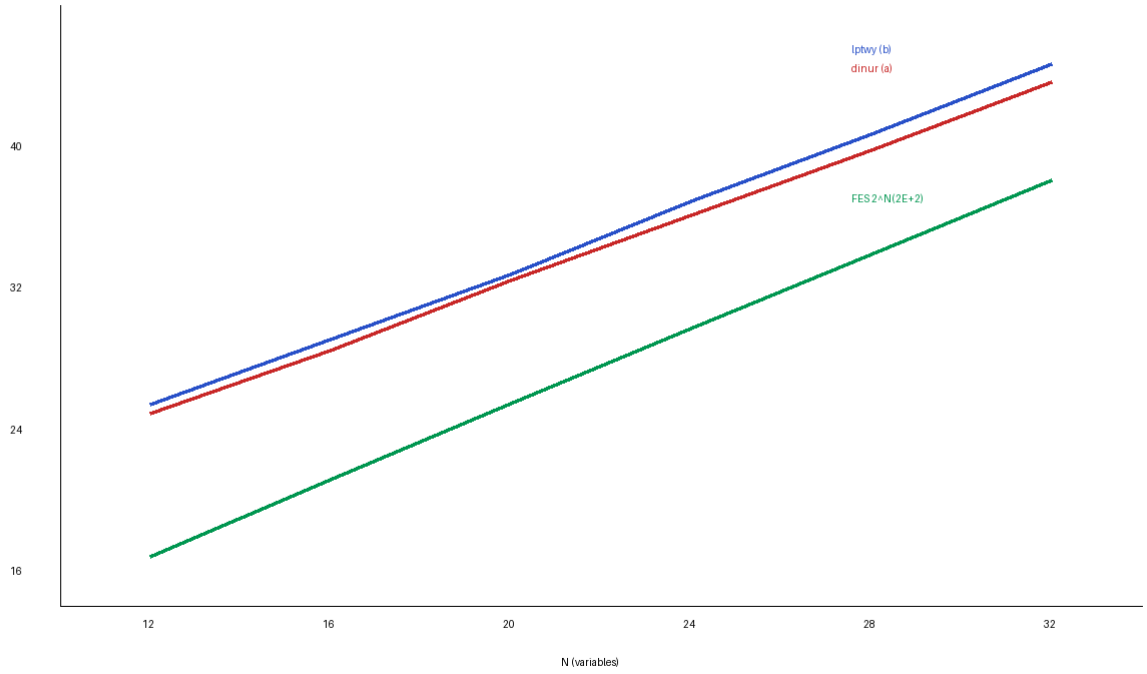
Gap to matched rho (AGENTS.md section 5a)

No cross-method row exists: there are no measured group-operation or unpriced native-operation rows, so pricing at 0, 0.1 and 1 GAE per operation is not applicable and no ranking is made. The only gap quoted is the asymptotic distance above.

Deviations

Cards 02 to 09 were refused by the executor with "card does not lint: ... names repository file target/blindspots-03/bin/b03 (b05) that binds does not pin" (tool exit 2, nothing ran). Cause: the cards were authored when the binary did not exist; once card 01 built it, the lint treats it as an unpinned repo file. Re-running task_card.py check-relay on every card, including card 01, now reports CARD FAIL for the same reason. Per instructions the relays are reported, not repaired; the cards were not edited. The user's request to restart so the builds run was honoured (card 01 built both binaries), but the rest cannot proceed without a director-side card change (for example excluding build outputs from the pin rule or pinning them after the build). Executor relays are in raw/relay-1.txt to raw/relay-9.txt (card 01's relay is condensed; its full output is in raw/card-01-*.log).

Caveats

Counted model only; host is shared 4-vCPU at isolation level L0; toy scale (N <= 32); the sensitivity control disagrees with the frozen text: PROTOCOL.md requires the dinur count to move by less than 1% across E = 17, 33, 64, but the instrument measured 1.36% (E=33) and 3.99% (E=64) and its code passes at rel_d < 0.05. Read against the frozen 1% the control fails; read against the code's 5% it passes. This code/protocol mismatch is reported, not resolved; it is a second reason the verdict is INCONCLUSIVE and needs a director decision. The built binaries remain in target/blindspots-03/bin because card 09, which removes them, did not run. No speed result, so no benchmarks/ entry.

Item 03 counted-model log2 cost, S_3 cells (verdict INCONCLUSIVE: real runs not executed)